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Temperature-based structural health monitoring baseline for long-span bridges

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ABSTRACT

A core prerequisite of an effective structural health monitoring (SHM) system is the development and characterization of a baseline response that is sensitive to meaningful changes in the structural system, and insensitive to normal operational changes. Such a baseline allows the use of detected changes to drive proactive maintenance and preservation interventions, or more refined assessment approaches, to ensure the on-going safety, serviceability, and durability of the structure. The approach developed as part of this research utilizes the relationship between temperature changes and the resulting strains and displacements of the structure to create a unique numerical and graphical baseline within an SHM framework. Evaluation of the method was performed through benchmark studies along with long-term monitoring data from a long-span steel tied arch bridge. The benchmark studies and field measurements illustrate that the nonlinear relationship between temperature, local mechanical strains, and global displacements results in a near-flat surface when plotted in 3D space. The bounds and the orientation (angle) of these surfaces are unique for each location and insensitive to normal operational changes in behavior. More importantly, a numerical sensitivity study was performed which indicated the surfaces are sensitive to a series of realistic scenarios which would result in meaningful changes in the performance of the structure. In addition, a comparison with a vibration-based SHM approach was also carried out, and the results indicated that the temperature-based approach was more sensitive for the scenarios examined.

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1. Introduction

Long-span bridges support vital arteries for national transportation systems, serve as lifelines across waterways and otherwise impassable terrain, and play a substantial environmental, social, and economic role in their respective regions. The majority of the long-span bridges within the U.S. were constructed during the first half of the 20th Century and thus are approaching their initially envisioned service lives. However, due to political, historical, and financial constraints, such structures have proven difficult to replace, and thus proactive approaches to preserve and renew these critical assets are becoming more relevant.

The state-of-the-art in long-span bridge assessment (to inform preservation activities) is continually being advanced with new developments in sensor technologies, information/communication technologies, and various data processing, visualization, and mining algorithms to aid in data interpretation. In recent years, utilizing field measurements within the paradigm of structural identification [1] has become commonplace when assessing the vulnerability or diagnosing performance problems of signature bridges [2–7]. Although not yet as common, utilizing field measurements to track longer-term performances through structural health monitoring (SHM) applications is beginning to enter the practice (as evidenced by on-going signature bridge projects including the Tappan Zee, Geothals, and Bayonne Bridges, among others) – delayed no doubt by a series of early applications that failed to live up to their billing.

One of the most common methods for SHM of long-span bridges is an ambient vibration-based approach (VBSHM) [8–15]. This method provides an overall characterization through tracking the modal parameters of the structure, and while it has enjoyed significant attention over the last several decades, it has many widely recognized drawbacks [16]. First, readily tracing changes in modal parameters to their root causes is difficult. Second, the relative insensitivity of modal parameters to local structural changes is challenging as such changes may be masked by varying environmental conditions [17]. Additional weaknesses include the

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unknown nature of the inputs that are assumed as wide banded white noise, predication on modal theory assumptions (linearity, stationary, etc.), and significant data processing and storage requirements. Some of these shortcomings are gradually being mitigated by advances in technology; however, others will persist as they are associated with fundamental assumptions that are implicit within the method itself.

As an alternative to vibration-based techniques, there is increasing attention being paid to the direct use of temperature and temperature induced responses for both structural identification [18,19] and SHM of long-span bridges [20–22]. Logistical advantages of temperature-based SHM (TBSHM) include large signal-to-noise ratios, low required sampling rates, and inexpensive sensing, data acquisition, data storage, and data transmission costs. As a result, TBSHM appears to have potential to overcome the current challenge of demonstrating attractive benefit-to-cost ratios to owners of long-span bridges.

The overarching goal of the research presented herein is to evaluate a novel three dimensional (3D) numerical and graphical TBSHM baseline for its potential to provide a reliable signature that is (a) insensitive to normal operational changes, and (b) highly sensitive to relevant and realistic damage scenarios. To accomplish this, simple benchmark numerical models were examined under several scenarios culminating in an application on a long-span tied-arch bridge. To place TBSHM in proper context, a comparison with VBSHM was carried out on a calibrated FE model of the tied-arch bridge in which several realistic damage scenarios were examined.

2. Concept and approach

TBSHM aims to track, characterize and ultimately identify and interpret changes to the relationship between responses (strains, displacements, and tilts) and the variations in temperature that induce them. The primary technical advantage of this approach over vibration-based methods is that the forcing function (i.e. temperature fluctuations) can be measured and thus a full transfer function (or input–output relationship) can be obtained (the underpinning assumptions of this transfer function are discussed below). In addition, TBSHM has several practical advantages such as (1) large signal to noise ratios, (2) extremely low power consumption due to relatively low sampling rates, and (3) relatively inexpensive sensing, data acquisition, and communication requirements.

To illustrate the concept, consider a simply supported beam with a longitudinal spring subjected to a uniform temperature change (Fig. 1). The mechanical strain (defined as the restrained portion of the strain (resulting from restrained displacement) that produces mechanical stress), ε_M, and unrestrained displacement (defined as the measured movement that does not produce mechanical strain), δ_U, as a function of the spring stiffness, k_S, coefficient of thermal expansion, α, uniform temperature change, ΔT, beam length, L, cross sectional area, A, and modulus of elasticity, E, is provided by Eqs. (1) and (2), respectively [23]. If the beam and the longitudinal spring are assumed to be linear, then a straight line results when ε_M, δ_U, and ΔT are plotted in 3D space. However, if an external elastic-perfectly plastic nonlinearity exists (e.g. k_s) or an internal elastic-perfectly plastic nonlinearity exists (e.g. EA), then the 3D plot becomes a surface.

$$\varepsilon_M = \frac{-k_S\alpha(\Delta T)L}{AE\left(1+\frac{k_S L}{AE}\right)} \tag{1}$$

$$\delta_U = \frac{\alpha(\Delta T)L}{1+\frac{k_S L}{AE}} \tag{2}$$

To illustrate the 3D surface behavior consider the simple beam model with a nonlinear k_s stiffness that is elastic-perfectly plastic (Fig. 2) with all other parameters (E, A, L, and α) linear. For this illustration the model is subjected to four uniform temperature changes (ΔT), with the initial condition of 0 °C. The first ΔT is an increase of 20 °C, which is the temperature change where the spring force reaches the bifurcation point (F_{slip}) and the ε_M, δ_U, and ΔT relationship becomes nonlinear. This is illustrated with point (a) in Fig. 3. A second ΔT increase equal to 20 °C is then applied to the model as shown by point (b). It is seen from Fig. 3 that the further expansion of the beam is not restrained (no increase of mechanical strain) by the spring since it is in the plastic range. The remaining two ΔT values applied were both decreases equal to −20 °C, which unloads the system and returns the overall temperature to 0 °C. These results are represented with points (c) and (d). As expected the behavior is nonlinear, but more importantly the ε_M, δ_U, and ΔT relationship maps a flat 3D surface as shown in the lower right plot of Fig. 3. To further illustrate this point a larger temperature time-history, consisting of 120 temperature cycles, was applied to the model following a typical seasonal temperature trend and the corresponding 3D surface plot was generated (Fig. 4).

The feasible limits (or bounds) of the 3D surface can also be illustrated from this model. Consider the limiting case of a rigid beam (i.e. infinite EA) and a nonlinear external spring (simulated by considering a range of stiffnesses). The resulting surface resides entirely in the temperature–displacement plane (Fig. 5) since no mechanical strain can develop. In the other limiting case of a rigid external spring and a nonlinear beam, the resulting surface resides entirely in the temperature-mechanical strain plane (Fig. 5) since all external displacement is restrained. The final case shown in Fig. 5 represents the surface for a beam with a finite elastic stiffness and a nonlinear external spring, which creates a surface that cuts across all three planes (Fig. 5).

The research reported herein, adopts these surfaces as a structural baseline and aims to establish their characteristics and sensitivities to various common damage scenarios. As illustrated through the simple example above, these surfaces are obtained

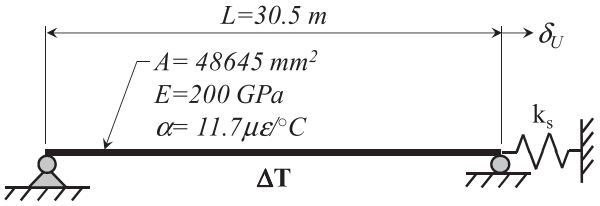


Fig. 1. Simple beam model.

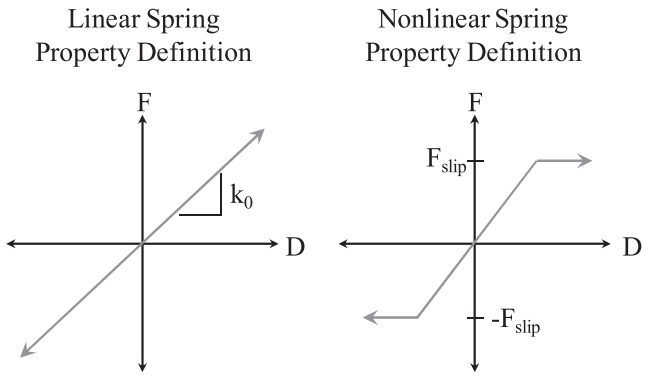


Fig. 2. Linear and nonlinear spring definitions.

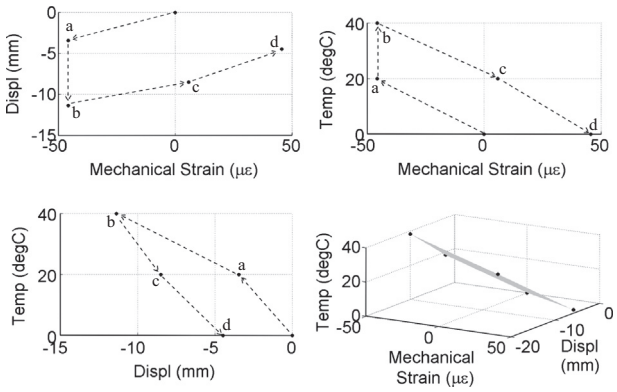


Fig. 3. Illustrative plot for ε_M, δ_U, and ΔT relationship (viewed at different angles).

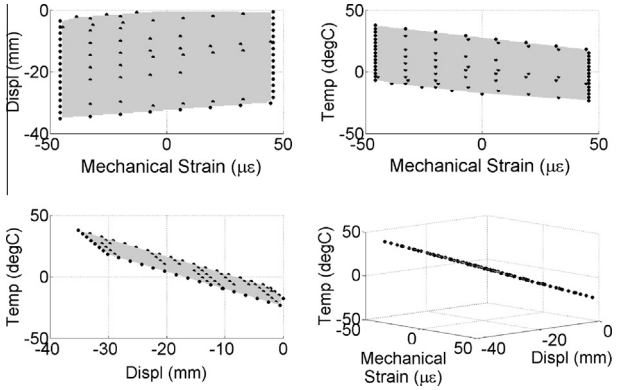


Fig. 4. Simple beam ε_M, δ_U, and ΔT relationship (viewed at different angles).

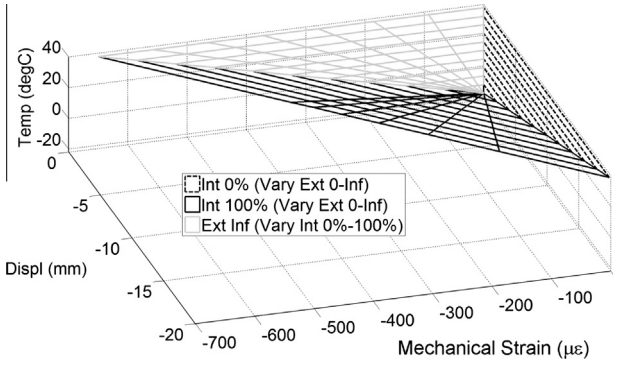


Fig. 5. Simple beam 3D surface limiting cases.

by measuring and then plotting the mechanical strains versus the displacement of movement mechanisms versus the change in temperature. These three quantities account for both the input (e.g. the temperature change) and the response of the structure – that is, whether the temperature induced deformation is accommodated through movement mechanisms, member distortion, or some combination.

3. Benchmark study

A comprehensive evaluation of the TBSHM method was conducted through various benchmark studies [23]. One of the studies was performed on a model which consisted of two simple beams connected with an internal spring (continuity condition) and supported with an external longitudinal spring (boundary condition) (Fig. 6). The objective of this benchmark study was to evaluate the influence (sensitivity) that boundary and continuity conditions have on the 3D relationship between mechanical strain (ε_M), unrestrained displacement (δ_U), and uniform temperature change (ΔT). This was achieved by mapping out the 3D baseline (ε_M versus δ_U versus ΔT) for various scenarios and then comparing them. Two scenarios were evaluated, with three cases per scenario, as shown in Table 1, that include different combinations of linear and nonlinear boundary/continuity conditions. The first case (Case A), of each scenario, evaluated the influence of changing the elastic stiffness (EA) of the beams. The second case (Case B) evaluated the influence of changes in the external spring response that included the influence of F_{slip} (Scenario 1) and k_{EXT} (Scenario 2). Similarly, the third case (Case C) evaluated the influence of changes in the internal spring response that included the influence of k_{INT} (Scenario 1) and F_{slip} (Scenario 2).

There are several limitations of the benchmark studies that are important for interpretation of the results. First, these studies only considered uniform temperature changes for structures where this demand resulted only in axial deformation and axial forces. Although the approach is equally applicable to cases where temperature changes result in bending forces (due, for example, to eccentricity (see [25]) this was not included in this illustrative study for the purposes of clarity. All material properties were assumed to remain constant with temperature change. In addition, the only nonlinear relationship considered was elastic-perfectly plastic behavior.

3.1. Scenario 1

The first scenario evaluated the 3D baseline sensitivity for a nonlinear (elastic–plastic) boundary condition and linear continuity condition (Table 1 – Scenario 1). Case A (varying EA) and Case C (varying k_{INT}) resulted in a change to the orientation (angle) of the 3D surface (Fig. 7a and Fig. 7c, respectively). This occurs since these parameters have a significant influence over the temperature at which the nonlinearity initiates. Essentially these parameters control how much temperature induced displacement is accommodated by the beam-spring system and how much is accommodated by the external spring when the system is responding elastically. This relationship defines the level of force a given temperature change induces, but has no influence over the δ_U associated with the yielding of the external spring (since this is only dependent on k_{EXT} and F_{slip}). Conversely, Case B (varying F_{slip} of the external spring) influenced the bounds of the 3D surface but not the orientation angle (Fig. 7b). This occurs since changing F_{slip} influences the δ_U, ΔT, and ε_M associated with the initiation of nonlinearity in the external spring in an identical manner. That is, if the slip force was reduced by half, then each of these response parameters would also be reduced by half (with their relative relationships remaining unchanged). As a result, the response surface stays at the same orientation, while expanding the bounds for a given temperature time history.

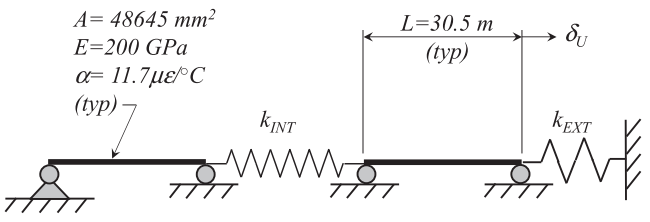


Fig. 6. Multi-beam benchmark study model.

Table 1
Benchmark study definitions.

Scenario	Boundary condition	Continuity condition	Case	*EA* (%)	Exterior spring (%)	Interior spring (%)
1	Nonlinear	Linear	A	25–400	100	100
	F_{slip} = 445 kN	k_{INT} = 263 kN/mm	B	100	25–400	100
	k_{EXT} = 175 kN/mm		C	100	100	25–400
2	Linear	Nonlinear	A	25–400	100	100
	k_{EXT} = 263 kN/mm	F_{slip} = 445 kN	B	100	25–400	100
		k_{INT} = 175 kN/mm	C	100	100	25–400

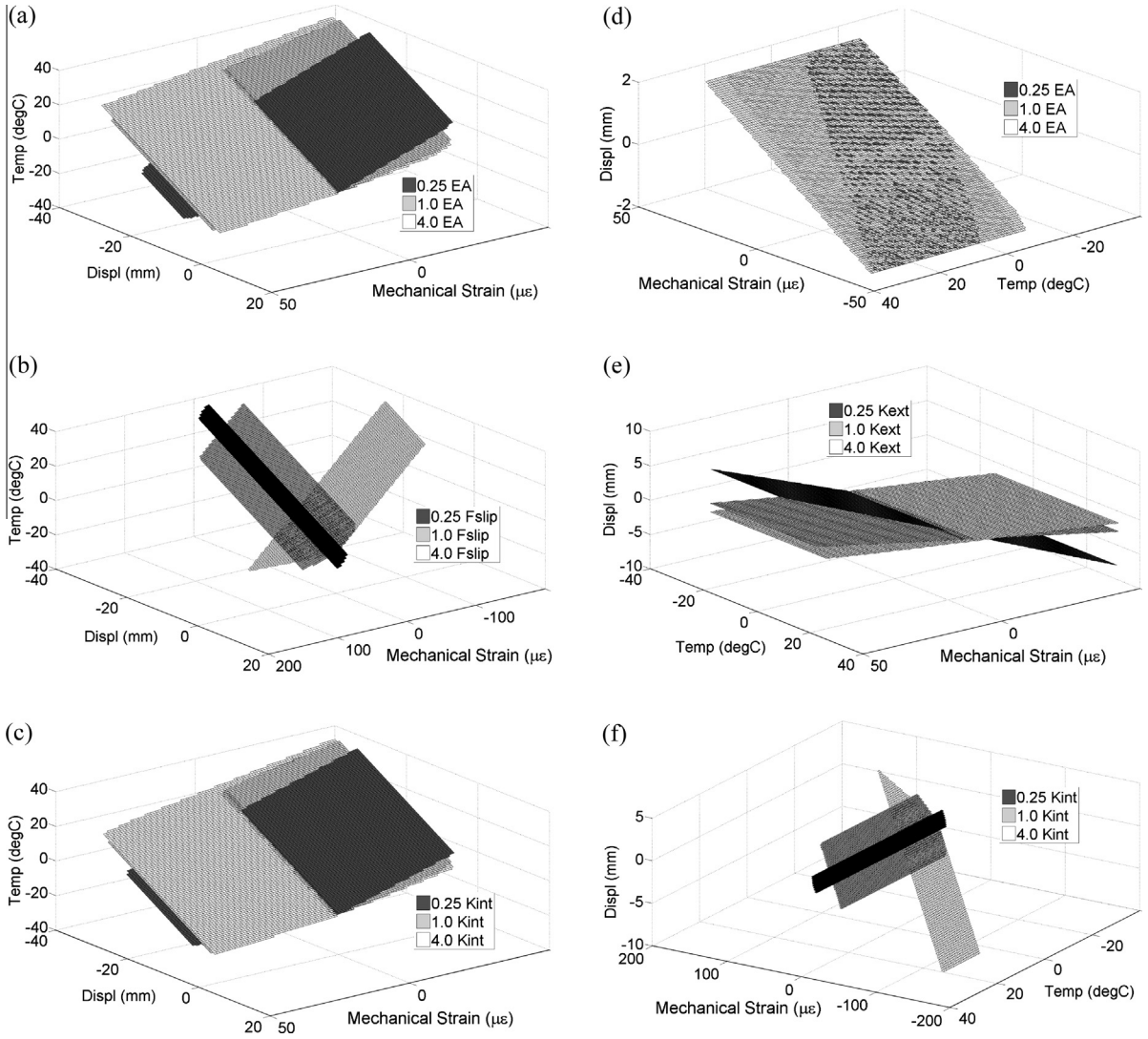


Fig. 7. Benchmark study plots for Scenario 1 (a) Case A, (b) Case B, and (c) Case C and Scenario 2 (d) Case A, (e) Case B, and (f) Case C.

3.2. Scenario 2

The second scenario evaluated the sensitivity of the 3D response surface to various parameters with a linear boundary condition and nonlinear (elastic–plastic) continuity condition (Table 1 – Scenario 2). Case A (varying *EA*) only affected the boundary configuration of the surface (i.e., the orientation of the boundaries of the surface) due to the constant slope from the ε_M versus δ_U relationship (Fig. 7d). However, Case B (varying k_{EXT}) resulted in changes to the orientation of the response surface due to alteration of the ε_M versus δ_U relationship for each k_{EXT} stiffness (Fig. 7e). In contrast, Case C (varying F_{slip} of the internal spring) resulted in a change in the bounds of the 3D surface, however no change in orientation (Fig. 7f). Similar to Case B in Scenario 1, this occurs since this parameter influences all three of the response variables (δ_U, ΔT, and ε_M) in a similar manner, thus their inter-relations remain unchanged.

3.3. Benchmark study summary

Based on the results and limitations described above, two general properties of the observed 3D response surfaces can be defined. First, the elastic response of the system defines the orientation of the plane, thus any changes to elastic properties cause the orientation to change. To understand how the orientation of the plot is defined, consider the response shown in Fig. 3. Once

nonlinearity initiates, the response moves along one of two possible paths. If the nonlinearity is external, then the response travels in the δ_U versus ΔT plane (constant ε_M) with a slope of αL, as any further temperature expansion is accommodated exclusively by the external nonlinearity (points a to b in Fig. 3). Conversely, if the nonlinearity is internal, then the response travels parallel to the ΔT axis as both δ_U and ε_M remain constant. Together with these paths, the orientation of the plane is defined by the elastic response of the system (e.g. point b to c in Fig. 3). Since the paths taken during nonlinear response are independent of both slip force and elastic properties, the only way to change the orientation of the response surface is to change the elastic parameters.

The second observation is that the slip force defines the bounds of the response surface, but does not have any influence over the orientation. The later point follows directly from the first observation, as the slip force has no influence over the elastic response. To examine the influence on bounds further, consider Fig. 7b, which shows the results of Scenario 1, Case B. As the slip force increases the response surface narrows as the level of nonlinear deformation decreases (for a given temperature time-history). If the slip force were to be increased further, eventually the response surface would collapse to a single line, which would coincide with the elastic response defined by Eqs. (1) and (2). Conversely, as the slip force drops, the surface bounds move in the opposite direction since the nonlinear deformation increases (for a given temperature time-history). If the slip force were to be further decreased to zero, then only nonlinear deformation would occur, and the plane would again collapse to a line. This time, the line would be associated with one of the two nonlinear paths described in the previous paragraph (depending on whether the nonlinearity is external or internal).

This mechanistic understanding of how various parameters influence the bounds and orientation of the response surface provides a basis to permit observed changes to be traced to their root cause. Further, such an understanding of expected behavior will also help assess whether or not underpinning assumptions (e.g. elastic–plastic response) are valid and allow for data quality checks.

4. TBSHM field study

To examine the sensitivity of the response surfaces in a realistic scenario, a field study was performed on the 168 m (550 ft) steel tied arch span (Fig. 8) of the Tacony-Palmyra Bridge (TPB), which was designed and constructed by Ralph Modjeski in 1929 [24]. The TPB primary structural system includes three chord members composed of built-up box shapes formed by riveted steel plates and angles. Using the original nomenclature from Modjeski, the arch is formed by upper and middle chords which are connected through vertical and diagonal members in and out of plane (Fig. 9). The lower chord (or tension-tie) runs horizontally below the roadway and is connected to the arch vertically through hangers, and horizontally through connections above the piers.



Fig. 8. TPB steel tied arch span (looking north).

The superstructure loads are transferred to the foundations through both fixed (NJ end) and expansion (PA end) bearings. Originally the expansion bearings were composed of nested steel rockers, which were replaced in 2001 due to "locking" that produced substantial build-up of forces at the NJ pier, resulting in cracking and other signs of distress. The replacement expansion bearings were a low profile friction-based assembly.

In addition to this primary movement system, the NJ end of the arch includes an additional movement mechanism within the middle chord that frames into the pier. This member is equipped with a slotted connection to accommodate movement longitudinally along this member (Fig. 10). In addition, the last panel of the upper and lower chords, at the NJ end (node U0), frame together with a pin-type connection, which allows for rotation in the plane of the arch (also shown in Fig. 10).

4.1. Instrumentation plan

The primary focus of the temperature-driven experiment was to adequately characterize the temperature-induced input–output relationships. The approach taken for identification of the input was to measure the temperature at all sensor locations along with ambient conditions at the site. The corresponding output measurements utilized two methods. The first was to measure the displacement at all movement mechanisms (e.g. bearings and slotted connections) due to their influence on force variation. The second method was to measure the mechanical strain of all primary structural members through different "slices" of the structure. Therefore, various free bodies of the structure could be examined to assess both data quality and the restrained temperature strains. To allow for bi-directional bending effects to be decoupled from axial forces, each measurement location (shown in Fig. 11) consisted of four vibrating wire strain gages located near each corner of the box members. Additional details on the design of the instrumentation plan can be found in [25].

4.2. Equipment and installation

The instrumentation plan (Fig. 11) for the TPB arch was installed in December 2010 and included the following equipment.

- 56 vibrating wire (VW) strain gages (Geokon, model 4000)
- 2 VW displacement gages (Geokon, model 4435)

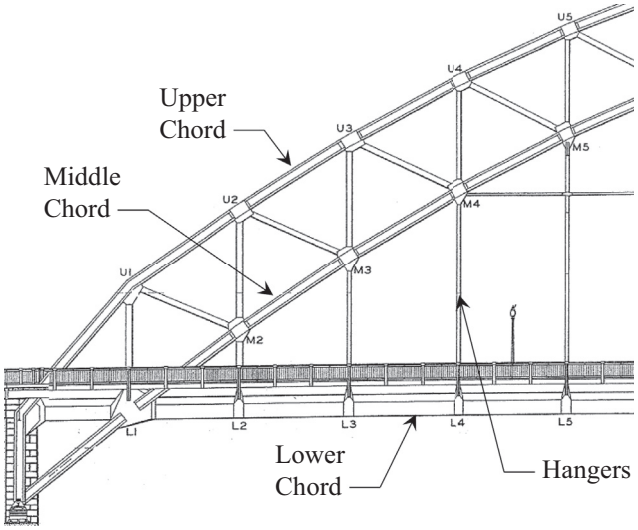


Fig. 9. TPB steel tied arch span elevation (PA end).

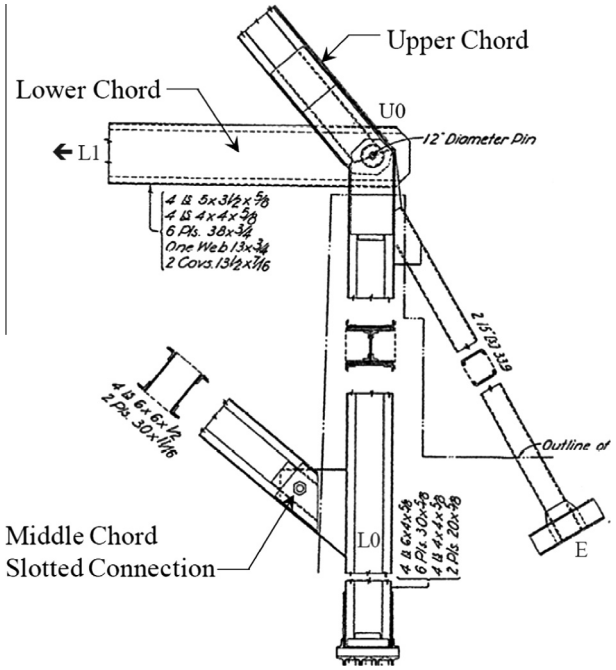


Fig. 10. TPB NJ (east) end connection detail.

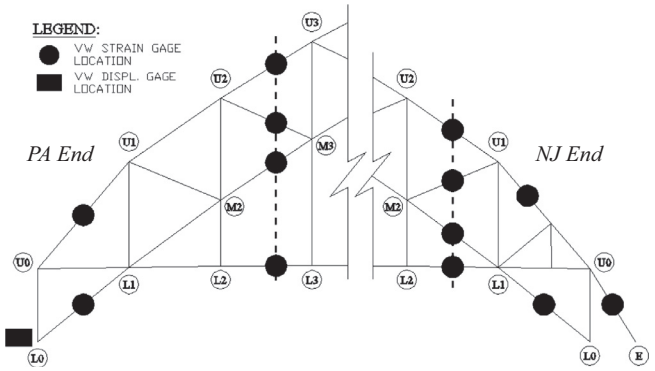


Fig. 11. TPB arch span instrumentation layout along the upstream and downstream side (looking north).

- 58 thermistors (Geokon)
- 2 data acquisition systems located at each end of the arch (Campbell Scientific, model CR1000)
- 1 weather station (Columbia, Orion Weather Station)

The full installation was completed in eight days with two people and no traffic disruption. All sensors were installed by climbing the structure with standard fall protection equipment. The data acquisition systems at each end of the span included cellular modems for remote connectivity. The sampling rate was set to 3 min to allow for minimal data storage requirements and easy data transmission.

The VW strain gages were shielded from direct solar radiation to avoid differential temperature variation of the sensor compared to the member. On the TPB arch span the majority of the gages were installed inside the box-shaped members. At locations where inside access was restricted, PVC covers were installed. Quality-control experiments were conducted on the arch span by using an infrared camera to ensure gage temperatures were the same as member surface temperatures [23].

4.3. Direct data interpretation

Yarnold et al. [25] and Yarnold [23] present a detailed direct analysis of the obtained data (i.e. prior to any model-experiment correlation). The primary conclusions from the analysis include:

- *Bearing behavior:* A nonlinear "stick–slip" movement mechanism was characterized using both displacement measurements of the expansion bearings as well as strain measurements of the members framing into the bearing (Fig. 12).
- *Slotted connection behavior:* Negligible axial strains were measured at the middle chord slotted connections, which indicated it was functioning well.
- *Response magnitudes:* Typical daily and seasonal strain and displacement magnitudes were directly obtained. Daily mechanical strains ranged from 20 με to 250 με with displacements ranging from 5 mm to 20 mm.
- *Equilibrium:* The intrinsic force variations along the global load path were verified with equilibrium through free body slices of the structure during near steady-state conditions.

4.4. Baseline measurement

As discussed earlier, the research focused on evaluation of the 3D response surface for its potential to provide a reliable signature. To form the response surfaces from field data, two data processing measures were applied to minimize local effects associated with temperature gradients and local bending. The first was to use evening only raw data which reduces the daily solar radiation effects and provides near steady-state behavior of the structure. The second approach was to average the four strain measurements taken at each cross-section to develop estimates of axial forces. This averaging was also used for the temperature thermistor measurements at each cross-section.

Using these approaches, response surfaces of the TPB arch were developed using three years of data. The obtained surfaces, which were formed through using strains from each location along the arch span together with displacement of the expansion bearing, were distinct for each strain location (unique bounds and orientation). Fig. 13 illustrates an example based on a one year dataset that includes the strain at the PA end lower chord member (L2L3) and displacement of the expansion bearing. A comprehensive reporting of all obtained response surfaces can be found in Yarnold [23]. The consistency of the obtained responses surfaces with those generated previously indicates that the assumptions that underpinned the benchmark studies (e.g. elastic-perfectly plastic nonlinear behavior) are also valid for the TPB arch span (due to the movement mechanism being a friction-based system).

5. Model-based comparison of TBSHM and VBSHM

For a measured baseline to be valuable within a SHM system it must be sensitive to relevant parameters, thus able to identify important changes in structural response. To examine how sensitive the response surfaces were to common damage scenarios for the TBP arch span, a series of numerical studies were carried out. These studies employed an FE model of the arch span that was calibrated to the measured responses described in the previous section, complete with the nonlinear boundary condition [25]. A series of damage scenarios (described below) were then defined based on historic records from the TPB, and simulated to assess the sensitivity of the response surfaces.

To provide some context to this study, a similar study was carried out using an ambient vibration-based (VB) SHM approach. For this study, the model employed was the same as for the TBSHM

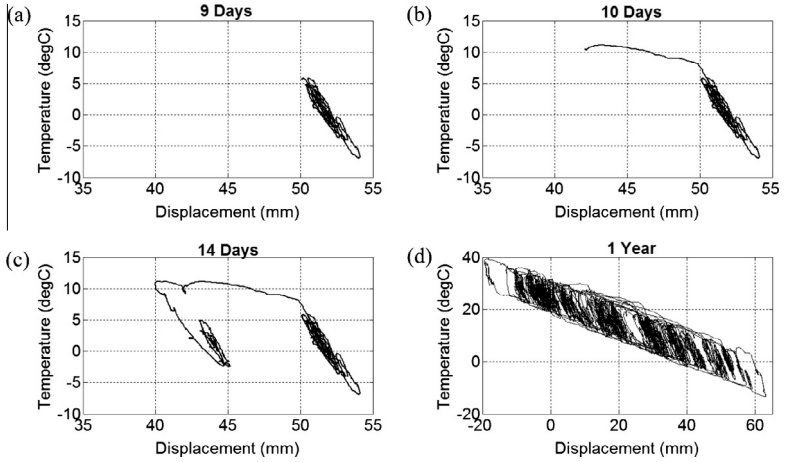


Fig. 12. Temperature versus displacement relationship over (a) 9 days, (b) 10 days, (c) 14 days, and (d) 1 year.

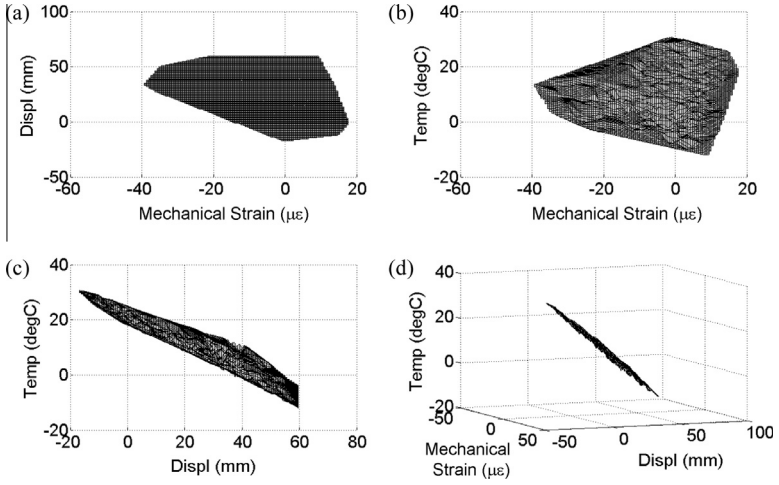


Fig. 13. West end lower chord (L2L3) 3D surface plot of evening only data.

study except its parameters were identified through comparison with an ambient vibration test as opposed to the temperature-driven response [25]. For each damage scenario investigated, the relative sensitivities of both the response surface and the modal parameters (frequencies and mode shapes) were obtained.

5.1. Simulation modeling and baseline analyses

5.1.1. Temperature-based SHM

To assess the sensitivity of the temperature-driven response surfaces, two primary performance metrics were utilized. The first employed a direct comparison of the magnitudes of mechanical strains and displacements, which was used to assess the changes in the response surface bounds. The second metric utilized a unit vector normal to the best-fit plane generated for each surface. The direction cosines were obtained from these vectors and the variation of these values was used to assess the changes in surface angle.

To conduct the numerical study a previously calibrated nonlinear 3D FE model was utilized. The element-level model was constructed of 3300 frame, 1750 shell, and 280 link elements based on existing documentation and information obtained directly from the measured data. This included an elastic-perfectly plastic boundary conditions to simulate the "stick–slip" friction bearing at the NJ-end (east end) of the arch span.

Calibration of the FE model was performed with temperature-based data sets obtained from one-day time windows. A total of

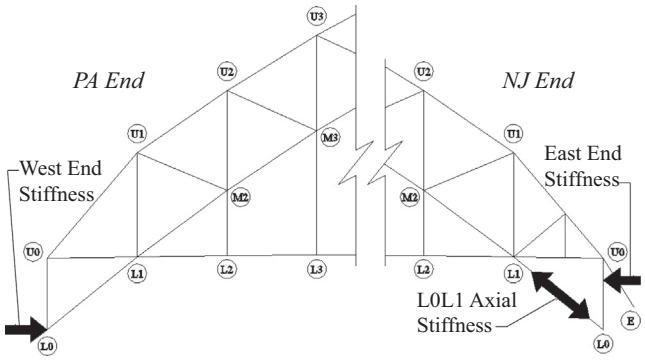


Fig. 14. Parameter locations for FE model calibration.

eight parameters associated with the nonlinear and linear boundary conditions as well as the effective stiffness of the slotted connection were selected for identification based on a series of sensitivity studies. Fig. 14 illustrates the spatial location of the parameters used within the model updating process. For parameter identification an objective function (which included model-experiment comparison of local strains and global longitudinal displacements) was minimized using a gradient-based optimization approach to reduce the discrepancy between the

measured and simulated responses. For the TPB application the objective function shown in Eq. (3), was employed to quantify the normalized error terms (R) at each time step (i) for each member (j), where n is the total number of time steps, and m is the total number of members evaluated. The error term (R) was defined as the straight line distance between the measured and FE model strain versus displacement results (see Yarnold [23] and Yarnold et al. [25] for additional details).

$$Obj\ Fcn = \sum_{j=1}^{m}\sum_{i=1}^{n} R(i,j)^2 \tag{3}$$

The model updating was carried out using the Strand7 [27] application programming interface (API) that allows seamless integration with Matlab [28]. To guard against the identification of local minima, several starting values were used for the parameter set, and they were found to converge to the same updated values [23]. Table 2. provides a summary of the identified parameters. Further details with regard to calibration of the FE model can be found in Yarnold et al. [25].

The approach for the baseline study was to first simulate the temperature-driven response surfaces for "nominal" (or pre-event) behavior from the updated FE model. This was performed by applying temperature inputs ("loads") to the model. These temperature inputs were generated based on the field-measured temperature time histories and included both typical daily and seasonal variations. To generate the response surfaces, a nonlinear time-step analysis was performed for the temperature inputs and the responses (mechanical strains and displacements) were extracted at each time step.

After the "normal" baseline behavior was established, the model was modified to simulate a series of damage scenarios (discussed below) and reanalyzed to generate the associated temperature/displacement/mechanical strain time-history responses. For consistency, the same temperature inputs were applied to the damaged models as used for the analysis of the undamaged, baseline model.

5.1.2. Vibration-based SHM

A vibration-based baseline was also evaluated to identify the sensitivity to the same damage scenarios employed for the evaluation of TBSHM. Two metrics were utilized to evaluate the sensitivity of VBSHM and included both changes in natural frequencies and mode shapes (quantified through the modal assurance criterion (MAC)). For this study the first eight global modes were considered.

Similar to the temperature-based method, this study was carried out on a previously calibrated 3D FE model from an ambient vibration test of the arch span [25]. In September 2011 an ambient vibration test was carried out using 48 uniaxial accelerometers located to identify the primary modes in the vertical, transverse, and longitudinal directions. Full details of the experiment and data processing can be found in Yarnold [23].

The processed results from the vibration test were utilized for conventional FE model calibration matching natural frequencies and MAC values. Eight similar parameters were selected for identification based on a series of sensitivity studies (Fig. 14). To identify the parameters, an objective function was minimized using a gradient-based optimization approach. The objective function is provided in Eq. (4) where f_{exp} and f_{an} are the experimental and analytical frequencies, respectively, and ϕ_{exp} and ϕ_{an} are the experimental and analytical modes shapes, respectively. Pairing of experimental and analytical modes was achieved using modified modal assurance criteria (MAC) values [29] and was verified through manual examination of paired modes throughout the updating process. Table 2 provides a summary of the identified parameters. Further details with regard to the calibration and discussion of results can be found in [25].

$$Obj\ Fcn = \sum_{i=1}^{n}\left[100\left(\frac{|f_{exp}(i) - f_{an}(i)|}{f_{exp}(i)}\right) + 100(1 - \mathrm{MAC}(\phi_{an}(i), \phi_{exp}(i))\right]^2 \tag{4}$$

The modal parameters (frequencies and mode shapes) associated with the "baseline" (or "undamaged") structure were obtained through eigen analysis of the updated FE model. The model was then modified to simulate the same series of damage scenarios referenced in the previous section, and the associated modal parameters were recomputed.

5.2. Overview of damage scenarios

To identify realistic damage scenarios, the historic records of the TPB were reviewed and three past events were selected. The first scenario (A) was bearing failure at the PA end (west end) of the structure. This was considered a feasible scenario due to the fact that the existing documentation indicates the original expansion bearings failed (locked-up) causing rapid deterioration along the substructures at the "fixed" end. Simulation of this scenario was performed by increased slip force and elastic stiffness of the bearings by a factor of five.

The second scenario (B) evaluated was the failure of the slotted connection at the NJ end (east end) of the structure located within the middle chord members L0L1 (Fig. 10). Breakdown (or failure) of this movement mechanism was considered feasible due to the steel-on-steel movable components. These types of connections can degrade over time if not properly maintained. Simulation was performed by increasing the continuity condition stiffness from 0% to 100%.

The last scenario (C) evaluated was failure to the lower chord (L4L5 location, downstream side). This was considered a feasible

Table 2
TBSI and AVSI FE model calibration results.

Parameter		Temperature-based			Vibration-based
		ΔF	Δd	Elastic stiffness	Elastic stiffness
West End Long. Stiffness	US	1160 kN	5 mm	230 kN/mm	130 kN/mm
	DS	1200 kN	5 mm	240 kN/mm	130 kN/mm
East End Add'l Long. Stiffness	US	–	–	190 kN/mm	1030 kN/mm
	DS	–	–	280 kN/mm	1470 kN/mm
L0L1 Axial Stiffness	US	–	–	0%	65%
	DS	–	–	0%	72%
Steel Modulus of Elasticity	–	–	–	100%[a]	99%
Steel and Concrete Mass	–	–	–	100%[a]	102%

[a] Value held constant during optimization.

Table 3
Magnitude changes of mechanical strains and displacements used to assess the changes in the surface bounds.

Damage scenario		Displ.	Strain (%)	Location
A	Avg	31%	252	–
	Max	–	375	U2U3 (PA)
	Min	–	0	L0L1 (NJ)
B	Avg	3%	114	–
	Max	–	334	U1U2 (NJ)
	Min	–	1	U2U3 (PA)
C	Avg	2%	5	–
	Max	–	40	L2L3 (NJ)
	Min	–	0	L0L1 (NJ)

scenario for several reasons. The lower chord could change stiffness due to ship impact, fracture, etc. There was a documented ship impact of the lower chord that required an emergency repair and shut down the bridge temporarily in 1988. In this scenario simulation was performed by reducing the continuity condition stiffness from 100% to 0% through incorporating an axial release within the element.

5.3. Discussion of results

A summary of the performance metrics for both the TBSHM and VBSHM approaches are provided in Tables 3–5. Overall, for the damage scenarios evaluated, temperature-based 3D surface baseline criterion was more sensitive to parameter variation compared to the vibration-based indices. The advantage of the temperature-based approach is that local (strains) and global (displacement) measurements are obtained. However, the vibration-based performance measures (frequencies and MAC values) are greatly smeared global metrics and therefore difficult to identify localized effects. The results from the TBSHM and VBSHM studies are briefly discussed below.

5.3.1. TBSHM results

Overall, the temperature baseline was sensitive to the three damage scenarios investigated. Damage scenario A (bearing failure) illustrated the increase in slip force has an effect on the bounds of the response surface, since it reduced the level of nonlinear deformation (Table 3). This was analogous to the results from Benchmark Scenario 1 (Case B). Fig. 15 graphically illustrates the typical bounds change in the response surfaces at location L0L1 (PA end). Since this response is global in nature (i.e. it influences the total force that is generated in the system due to temperature changes) the response surfaces were similarly impacted. While the orientation angle changed slightly due to the increased elastic stiffness of the bearing, this change was essentially negligible (Table 4).

Considering scenario B (slotted connection failure – NJ end), the response surfaces showed a change in orientation since the removal of the release was equivalent to increasing the elastic stiffness of this member. This was consistent with the findings from Benchmark Scenario 1 (Case A and C). However, since this parameter's influence over elastic response decreased as the distance from the connection increased, only the response surfaces obtained from strain readings in the vicinity of the connection showed appreciable

Table 5
Changes in natural frequency and MAC values to assess ambient vibration.

Damage scenario		Freq. (%)	MAC
A	Avg	2	0.99
	Max	6	1.00
	Min	0	0.94
B	Avg	0	1.00
	Max	1	1.00
	Min	0	1.00
C	Avg	1	0.99
	Max	1	1.00
	Min	1	0.98

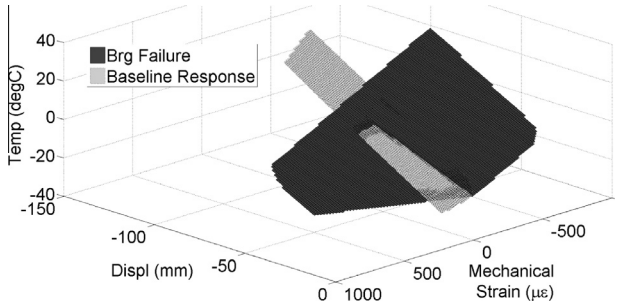


Fig. 15. PA (west) end middle chord (L0L1) 3D plots before and after bearing failure.

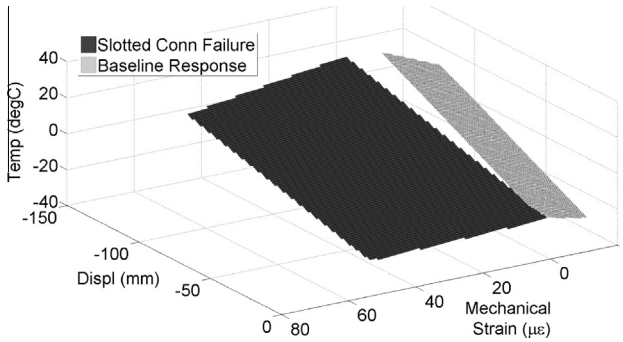


Fig. 16. NJ (east) end upper chord (U1U2) plots before and after slotted connection failure.

Table 4
Changes in direction cosines to assess changes in surface angle.

Damage scenario	Direction cosine	PA side (west)			NS side (east)		
		L0L1 (%)	L2L3 (%)	U2U3 (%)	L1M2 (%)	U1U2 (%)	L0L1
A	Strain axis	0	0	0	0	0	–
	Displ. axis	0	0	0	0	0	–
	Temp. axis	0	0	0	0	0	–
B	Strain axis	10	12	8	125	84	Inf
	Displ. axis	0	0	0	0	0	Inf
	Temp. axis	0	1	0	9	12	Inf
C	Strain axis	19	90	10	13	29	–
	Displ. axis	0	0	0	0	0	–
	Temp. axis	1	0	1	0	7	–

changes in orientation (Table 4). Fig. 16 graphically illustrates the change in the response surface at location U1U2 (NJ end).

For scenario C (bottom chord failure – PA end) a response similar to that of the slotted connection failure was observed. Since this damage scenario influenced the elastic response locally, response surfaces derived from strain measurements in the vicinity of the damage show significant changes in their orientation. Conversely, response surfaces obtained further away from the damage location showed small/negligible changes (Table 4).

5.3.2. VBSHM results

In contrast to the sensitivities of the temperature-based response surfaces, the modal parameters experienced relatively small changes due to each damage scenario (Table 5). It is important to emphasize that this observation is only valid for the scenarios investigated. While these scenarios were selected due to their past occurrence, due to their rather significant influence over how temperature forces flow longitudinally, and their limited influence over global stiffness, it is not surprising that the TBSHM showed more sensitivity. Vibration-based damage detection (VBDD) is an evolving field that hopes to identify localized issues through vibration measurements. However, VBDD has only been successful in a laboratory or controlled field settings and has not been proven for operational civil structures [16].

6. Conclusions and recommendations

The primary contribution of this paper is the development and evaluation of a novel 3D numerical and graphical temperature-driven baseline for long-span bridge SHM systems. At the heart of every SHM system is a baseline for identifying changes that indicate various performances (strength/safety, serviceability, functionality, etc.) are degrading, which would trigger proactive interventions to ensure such performances are not fully compromised. The baseline implemented must be reliable and sensitive for detection of relevant parameter deviations (structural changes), but relatively insensitive to normal operational changes. The 3D relationship between local axial mechanical strain, global unrestrained displacement, and uniform temperature variation has shown the ability to meet these criteria through a series of benchmark studies, and a field monitoring and simulation study conducted on a long-span tied-arch bridge. The following specific conclusions and recommendations were drawn from the study.

1. The temperature-based 3D surface baseline orientation (angle) and bounds are controlled by the elastic and inelastic properties of the structure, respectively. The surface orientation is defined by the elastic response of the system, thus any changes to elastic properties cause the orientation of the surface to change. The bounds of the 3D surface baseline are defined by the inelastic slip force, but do not have any influence over the surface orientation. This information allows for the 3D surfaces to be used as a diagnostic tool for faster and more refined data interpretation.
2. For large scale structural systems, localized effects can be identified by changes to the 3D baseline; however, the local measurement component needs to be within the vicinity of the damage location. Therefore, it is recommended to instrument multiple cross-sections along the primary load path. To identify the required spatial resolution of these instrumentation slices, an FE modeling study that examines a series of realistic damage scenarios is recommended.
3. Global changes of a large scale structure (associated with the performance of movement systems) can be readily identified through changes to the 3D baseline since they affect the surfaces generated at all locations.
4. The 3D baseline surface has shown to be highly sensitive to realistic damage scenarios for the tied-arch structure examined. The arch bridge bearing failure scenario illustrated a substantial impact to the surface bounds which was quantified through the average changes in strains and displacement equal to 250% and 30%, respectively. The middle chord connection failure and lower chord failure scenarios demonstrated the orientation change in the surfaces through direction cosine changes reaching 125% and 90%, respectively.
5. A vibration-based baseline (based on natural frequencies and mode shapes) was found to be significantly less sensitive for the scenarios examined. The changes in natural frequencies ranged from 0% to 6% with negligible changes to the MAC values.
6. Due to the lower performance requirements of the data acquisition equipment (sampling rates and synchronization), minimal data processing and communication costs, and negligible data storage, the TBSHM approach is more economical than the VBSHM approach.
7. The primary drawback of the TBSHM approach (compared with the VBSHM approach) is associated with both implementation time (since large temperature swings are required to generate the baseline) and alert time, since certain damage scenarios may also require large temperature swings to diagnose. Quantifying these shortcomings for a series of structures and realistic damage scenarios is currently being studied by the authors.

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